

IMPLEMENTATION OF DATA STRUCTURES IN R**AIM:**

To implement the data structures in R programming.

PROCEDURE:

1. Define and Create Data Structures – Create different data structures, including vectors, lists, matrices, and data frames, by assigning appropriate values.
2. Use Functions to Initialize Data – Utilize functions such as `c()`, `seq()`, `rep()`, `list()`, `matrix()`, and `data.frame()` to define structured data.
3. Execute the Program – Run the R script in an R environment like RStudio to process the data structures.
4. Verify Output – Print the results to confirm the correct implementation of vectors, lists, matrices, and data frames.

SOURCE CODE:

```
# Vector
```

```
a <- c(10, 20, 30, 40, 50)
```

```
print(a)
```

```
b <- seq(1, 10, 1)
```

```
print(b)
```

```
c <- rep("MCA", 4)
```

```
print(c)
```

```
# List
```

```
a <- list("MCA", 35, 1:8, 56.3, month.abb)
```

```
print(a)
```

Matrix

```
a <- matrix(data = 1:10, nrow = 3)
```

```
print(a)
```

Data frame

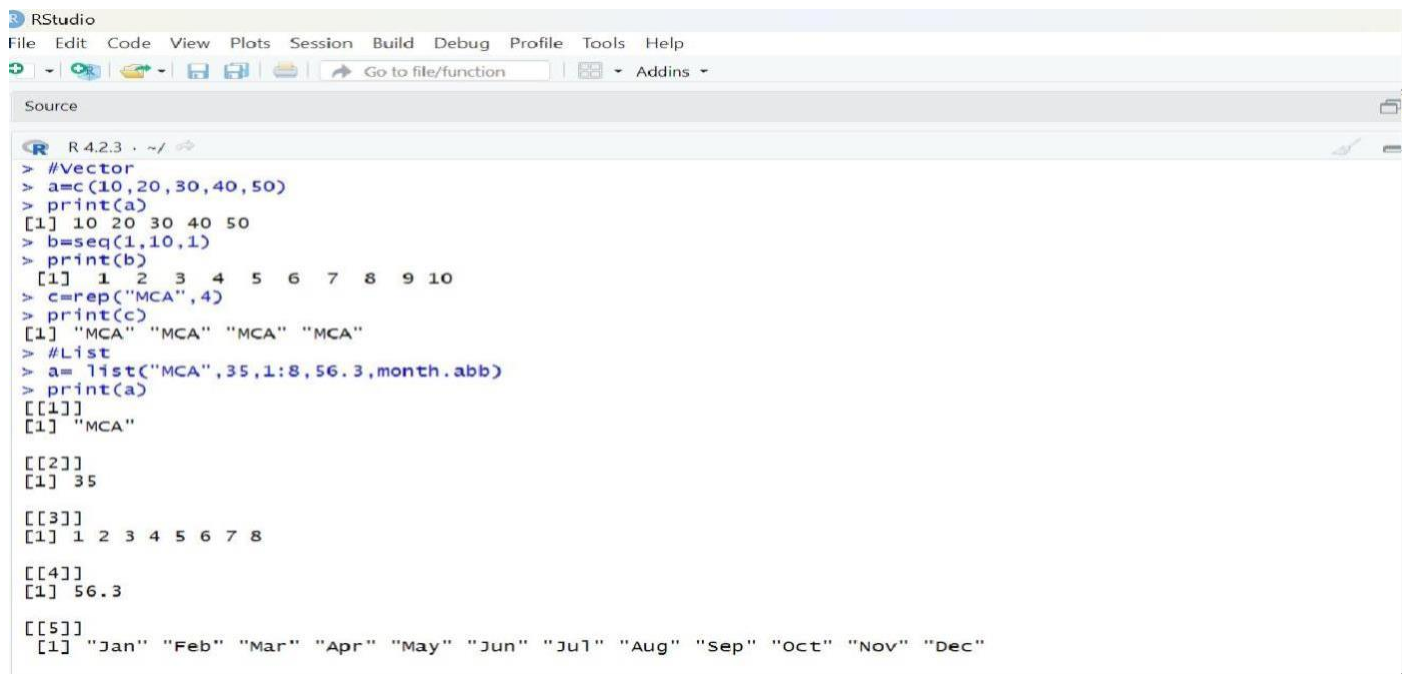
```
courses <- c("MCA", "MBA", "BCA", "B.COM")
```

```
students <- c(49, 60, 56, 43)
```

```
college <- data.frame(course_name = courses, students_count = students)
```

```
print(college)
```

OUTPUT:

A screenshot of the RStudio interface. The top menu bar includes File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, and Help. Below the menu is a toolbar with icons for running, saving, and other functions. The main window is titled 'Source' and contains the following R code:

```
> #Vector
> a=c(10,20,30,40,50)
> print(a)
[1] 10 20 30 40 50
> b=seq(1,10,1)
> print(b)
[1] 1 2 3 4 5 6 7 8 9 10
> c=rep("MCA",4)
> print(c)
[1] "MCA" "MCA" "MCA" "MCA"
> #List
> a= list("MCA",35,1:8,56.3,month.abb)
> print(a)
[[1]]
[1] "MCA"

[[2]]
[1] 35

[[3]]
[1] 1 2 3 4 5 6 7 8

[[4]]
[1] 56.3

[[5]]
[1] "Jan" "Feb" "Mar" "Apr" "May" "Jun" "Jul" "Aug" "Sep" "Oct" "Nov" "Dec"
```

The output of the code is displayed in the console area below the source editor.

```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins

Source

R 4.2.3 . ~/
[1] 35

[[3]]
[1] 1 2 3 4 5 6 7 8

[[4]]
[1] 56.3

[[5]]
[1] "Jan" "Feb" "Mar" "Apr" "May" "Jun" "Jul" "Aug" "Sep" "Oct" "Nov" "Dec"

> #Matrix
> a=matrix(data=1:10,nrow=3)
warning message:
In matrix(data = 1:10, nrow = 3) :
  data length [10] is not a sub-multiple or multiple of the number of rows [3]
> print(a)
     [,1] [,2] [,3] [,4]
[1,]    1    4    7   10
[2,]    2    5    8    1
[3,]    3    6    9    2
>
> #Data frame
> courses=c("MCA","MBA","BCA","B.COM")
> students=c(49,60,56,43)
> college=data.frame(course_name=courses,students_count=students)
> print(college)
  course_name students_count
1         MCA              49
2         MBA              60
3         BCA              56
4        B.COM              43
>
```

RESULT:

Thus, the data structures in R programming have been successfully implemented.